

All of Us Scientific Roadmap Refresh — Executive Summary

2026 Science Committee Working Group · Co-leads: Jordan Smoller & Melissa Haendel · Draft 2026-06-02

THE QUESTION

The Science Committee asked WG-1 to refresh the All of Us Scientific Roadmap against the Roadmap 2.0 six questions: what has changed since 2023; the largest data gaps; reproducibility expectations; new tools needed; what exposures and outcomes to capture; and **whether the 8 priority focus areas are still right**. Our charge is uniquely structural — only this WG can change the framework the other four 2026 WGs must fit inside.

THE EVIDENCE BASE

We analyzed **1,378 substantive AoU publications** and **12,899 workspace projects**, theme-tagged against a locked 19-theme taxonomy. We crosswalked themes onto the 8 focus areas and computed pre/post-2023 trends. We extracted **166 strategic priorities** from **29 NIH strategic plans** (NIH-Wide + 21 ICs) and built an alignment matrix. We compared AoU against **4 peer biobanks** (UK Biobank, FinnGen, MVP, China Kadoorie) using a same-tagger approach. Recommendations were scored against the 8-criterion Sci Com rubric Smoller established in November 2024.

HEADLINE FINDINGS

- The **8-area framework holds at the top level** but has three structural problems: a Methods/AI orphan (18% of pubs, no home), wearables orphaned despite being a slide-7 priority, and multi-omics invisible inside "Genetics & Biology."
- Strongest IC alignments:** NHGRI×Genomics (largest single matrix cell; AoU delivers Bold Prediction #9), NIMHD×SDOH (Goal 7 literally describes the AoU mandate), NLM×Methods (the only IC plan that explicitly names AoU).
- Biggest demand-side gaps:** NIEHS×Environment (6 IC priorities vs. 7 AoU pubs — the single largest gap); imaging across NCI/NEI/NIBIB (data-supply, not demand); multi-omics for NIDDK/NHLBI (the 2026 omics releases should close this mechanically).
- Peer-biobank comparison:** AoU is under-indexed ~4x on rare disease, 6x on multi-omics, 12x on imaging, 4x on aging/ADRD — but **leads peers 21x on ELSI, 5x on SDOH, 4.6x on PGx, 3x on methods**. Distinctive ≠ growing; these four over-indexes are AoU's defensible identity.
- Pan-NIH triangulation:** five plans articulate explicit pan-IC infrastructure demand (NIH-Wide Objective 1; NLM Goal 1; NHGRI Bold Predictions #5/#9; NIMHD Goal 7; NIH-Wide CCT4/CCT5) — empirical basis for an AoU-as-pan-NIH-infrastructure framing.
- Four slide-7 priorities are **already realized** (AI/foundation models 31 pubs; breakthrough therapeutics 16; wearables 70 — mature, not emerging; G×ESDOH 10, 0 pre-2023); four are **aspirational pending upstream investment** (multi-omics, imaging, microbiome, environmental).

TOP RECOMMENDATIONS

Recommendation	Rubric	WG / Action requested
Add a 9th focus area "Data, AI & Methods" (new) — absorbs the 18% Methods/AI corpus orphan, wearables, and the AI/foundation-model surge.	3.71	WG-1 — <i>endorse</i>
Rename FA8 (currently "Genetics & Biology") to "Genetics, Multi-omics & Biology" and commit to a proteomics-release roadmap (Olink/SomaScan scale).	3.38	WG-1 — <i>endorse + authorize</i>
Bundle FA reframes: FA4 ("Return of Results on Individual & Population Health") toward ROR implementation; FA2 ("Maternal & Child Health") to "Maternal Health" ; add explicit FA6 ("Environment") environmental-exposure linkage commitment.	3.00	WG-1 — <i>discuss + vote</i>
AoU Workbench AI co-scientist (cohort-builder + code-generator) — highest rubric mean across all five WGs.	3.88	WG-4 — <i>fund</i>
Cross-WG bias-in-translation framework stack (WG-2 normative, WG-3 imaging, WG-4 AI) — single AoU bias-reporting standard.	3.75	WG-2 (lead) / 3 / 4 — <i>commission</i>
Plain-language distinctive-evidence narrative + annual AoU Impact Report (ELSI 21x, SDOH 5x, PGx 4.6x, methods 3x).	3.60	WG-5 — <i>fund</i>

Action key. *Endorse* — formal Sci Com expression of support (non-binding for budget). *Authorize* — permission for staff to proceed with implementation, with resource commitment. *Vote* — recorded yes/no at the meeting, required to amend the Roadmap. *Fund* — allocation of program budget in the next cycle. *Commission* — formal delivery assignment to a WG, program team, or external partner.

Current 8 focus areas. FA1 Prevalent Common & Rare Conditions · FA2 Maternal & Child Health · FA3 Healthy Aging & Resilience · FA4 Return of Results on Individual & Population Health · FA5 Lifestyle, Substance & Behavioral Health · FA6 Environment · FA7 Health Equity · FA8 Genetics & Biology.

CAVEATS

- Dermatology over-representation (14.3% of corpus, peer index 7.8x) is an artifact** driven by 1–2 prolific groups; trend is declining (–3.9). Do not treat as a strategic priority.
- Peer-biobank counts use PubMed name-search fallback (same heuristic across peers); use the static *share index*, not trend differentials, for action-relevant comparisons.
- AoU pre-2023 baseline is tiny (157 pubs); apparent "growth" on cardiometabolic/mental health/cancer is a small-baseline artifact — read as "presence built," not "outpacing the field."

Full evidence base, per-IC briefs, and machine-readable companions at github.com/mellybelly/all-of-us